MINA

Developer guide and source map

Mediation, Integration and Automation

Version 1.0.3 - 104 activities, 11 groups, 103 functions and 12 shared connection types.

Repository navigation for every palette activity and core development feature. Source locations include verified search anchors and build-specific line numbers. Frontend, Python engine, native bridge and direct-export code paths are distinguished.

Use the linked contents or PDF bookmarks. Paths are relative to the repository root, not the installation directory. Edit maintained sources, add regression tests, rebuild and redistribute; do not patch generated archives as the permanent fix.

Contents

Developer guide: where to change MINA 7

Start: source map 8

Scheduler: source map 11

Call Sub Task: source map 14

End: source map 17

Read File: source map 20

Write File: source map 23

List Files: source map 26

Delete File: source map 29

Rename File: source map 32

Copy File: source map 35

File Poller: source map 38

FTP Get: source map 41

FTP Put: source map 43

FTP Delete: source map 45

FTP Directory: source map 47

FTP Change Directory: source map 49

SFTP Get: source map 51

SFTP Put: source map 53

SFTP Delete: source map 55

SFTP Directory: source map 57

SFTP Change Directory: source map 59

HTTP Listener: source map 61

HTTP Request: source map 64

Read Server-Sent Events: source map 66

WebSocket Exchange: source map 68

HTTP Write Preconditions: source map 70

HTTP Response: source map 72

REST API Receiver: source map 75

REST API Invoke: source map 78

SOAP Service: source map 81

SOAP Request Reply: source map 84

EMS Queue Receiver: source map 87

EMS Topic Subscriber: source map 90

EMS Queue Sender: source map 93

EMS Topic Publisher: source map 96

EMS Request Reply: source map 99

EMS Reply: source map 102

Get JMS Queue Message: source map 105

JMS Receive Message: source map 108

JMS Request Reply: source map 111

JMS Send Message: source map 114

Reply to JMS Message: source map 117

Wait for JMS Request: source map 120

Kafka Receive Message: source map 123

Kafka Send Message: source map 126

Kafka Get Messages: source map 129

Pub/Sub Subscriber: source map 132

Pub/Sub Publisher: source map 135

Dynamic Connection: source map 138

IDoc Acknowledgment: source map 141

IDoc Confirmation: source map 144

IDoc Converter: source map 147

IDoc Listener: source map 150

IDoc Parser: source map 153

IDoc Reader: source map 156

Post IDoc to SAP: source map 159

IDoc Renderer: source map 162

RFC BAPI Listener: source map 165

Invoke RFC BAPI in SAP: source map 168

Reply from RFC BAPI in SAP: source map 171

Read Table: source map 174

JDBC Create: source map 177

JDBC Update: source map 180

JDBC Query: source map 183

JDBC Truncate: source map 186

JDBC Delete: source map 189

Stored Procedure: source map 192

Dynamic SQL: source map 195

Snowflake Insert: source map 198

Snowflake Query: source map 201

Snowflake Update: source map 204

Snowflake Delete: source map 207

Snowflake Bulk Load: source map 210

AMQP Receive Message: source map 213

AMQP Get Message: source map 216

AMQP Send Message: source map 219

AMQP Dead Letter Message: source map 222

Parse XML: source map 225

Render XML: source map 227

Parse JSON: source map 230

Render JSON: source map 232

Parse Data: source map 234

Render Data: source map 236

Read Excel Workbook: source map 238

Mapper: source map 241

Transform: source map 243

Log: source map 246

Catch Exception: source map 249

Throw Exception: source map 252

Rethrow Exception: source map 255

Confirm Message: source map 258

Java Activity: source map 261

Python Invoke: source map 264

External Command: source map 267

Assign: source map 270

Checkpoint: source map 273

Deduplicate Message: source map 276

Message Receipt: source map 279

Replay Messages: source map 282

Aggregate Messages: source map 285

Split Records: source map 288

Sleep: source map 291

Get Shared Variable: source map 294

Set Shared Variable: source map 297

Tasks, sub-tasks, persistence and copy 300

Run, continuous events and Stop 301

Debug controls, test bench and Job Data 302

Input mapping, schema mapper and suggestions 304

Built-in and project functions 306

Canvas and mapper drag-and-drop 307

Groups: editor, graph boundaries and runtime state 308

If: source map 310

While True: source map 311

For Each: source map 312

Iterate: source map 313

Repeat Until True: source map 314

Repeat on Error Until True: source map 315

Scope: source map 316

None: source map 317

Critical Section: source map 318

JDBC Transaction: source map 319

Pick First: source map 320

Engine archives versus direct Python generation 321

Portable transformation tests and durable mediation 322

Logs, faults and observable execution data 323

Every built-in function: implementation index 324

Developer guide: where to change MINA

Developer guide / Start here

Architecture

This is a source-maintenance guide, separate from the activity user manual. Frontend code is React/TypeScript, core engine code is Python, and native connector work can execute in Java. Paths are relative to your repository checkout, not the installed program directory. Line numbers are regenerated from verified anchors and refer to this documentation build.

```
Palette / canvas -> ActivityEditor -> saved Project
Run API -> WorkflowRuntime.run -> execute_with_policy
 -> resolve_activity_config -> execute -> adapter / native bridge
 -> record_activity_output -> Run result / Debug Job Data

Raw Python export -> raw_python_files -> generated task modules
 -> raw_python_support (specialized helpers) -> adapters / bridge
```

The source tables tell you where to start, not that every option is fully implemented. Read the operational guide for known behavior gaps. A passing unit test or this code index does not establish provider certification.

How to navigate

Expand Developer guide > Activities for one page per palette activity. Features covers tasks, groups, Run, Debug, mapping, functions and drag-and-drop. Each source table provides a file:line plus a stable search anchor. Use your editor Go to File and Go to Line; if your checkout differs, search the anchor. The site deliberately does not expose source files or local filesystem URLs.

```
Example from repository root:
rg -n "def resolve_activity_config" backend/app/runtime.py
rg -n "file_activity" backend/app/raw_python_support/activities.py
```

Do not edit frontend/dist, installed app.asar, generated task archives or packaged engine copies as the permanent fix. Edit repository sources, regenerate/rebuild, and redistribute. Preserve saved activity IDs and config keys when changing labels.

Change checklist

1. Decide whether the change is visual, schema, runtime, transport or export behavior. 2. Follow the relevant source map. 3. Preserve existing project defaults/aliases or add an explicit migration. 4. Update runtime and direct-export implementations where they differ. 5. Add normal, error, cancellation and legacy-project tests. 6. Regenerate docs, build and test a newly packaged install.

```
cd backend
python -m unittest discover -s tests
cd ../frontend
npm run docs:build
node --test scripts/documentation.test.mjs
npm run build
```

Use a development environment with the project dependencies installed. Provider tests require isolated test infrastructure and licensed drivers. Never use production queues, write APIs or secrets in documentation fixtures.

Start: source map

Trace

Saved identity: type=start; config.operation=start. Start is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (start)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: start; select operation start	frontend/src/ActivityEditor.tsx:320 if (n.type === "start")
Engine start dispatch; operation start	backend/app/runtime.py:752 if activity.type == 'start':
Direct helper dispatch for start; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Task graph and interface schema	backend/app/models.py:108 class ProcessDefinition(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Start, select type=start and operation=start, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of start. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Scheduler: source map

Developer guide / Activities / Starters & Tasks

Trace

Saved identity: type=timer; config.operation=schedule. Scheduler is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (timer)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: timer; select operation schedule	frontend/src/ActivityEditor.tsx:334 if (n.type === "timer")
Engine timer dispatch; operation schedule	backend/app/runtime.py:758 if activity.type == 'timer':
Direct helper dispatch for timer; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Scheduler, select type=timer and operation=schedule, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of schedule. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Call Sub Task: source map

Trace

Saved identity: type=call_task; config.operation=call. Call Sub Task is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (call_task)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: call_task; select operation call	frontend/src/ActivityEditor.tsx:362 if (n.type === "call_task")
Engine call_task dispatch; operation call	backend/app/runtime.py:1160 if activity.type == 'call_task':
Direct helper dispatch for call_task; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Task graph and interface schema	backend/app/models.py:108 class ProcessDefinition(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Call Sub Task, select type=call_task and operation=call, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of call. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



End: source map

Trace

Saved identity: type=end; config.operation=end. End is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (end)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: end; select operation end	frontend/src/ActivityEditor.tsx:327 if (n.type === "end")
Engine end dispatch; operation end	backend/app/runtime.py:755 if activity.type == 'end':
Direct helper dispatch for end; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Task graph and interface schema	backend/app/models.py:108 class ProcessDefinition(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing End, select type=end and operation=end, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of end. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Read File: source map

Trace

Saved identity: type=file; config.operation=read. Read File is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation read	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation read	backend/app/raw_python_support/activities.py:50 def file_activity(
Text read implementation	backend/app/runtime.py:1100 content = path.read_text(
Binary read implementation	backend/app/runtime.py:1099 path.read_bytes()
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

Example: adding a streaming/size-limit option requires a palette default if needed, a Configuration field in activityContract, mapping exposure via runtimeMappableInputs, a check before read_text/read_bytes in WorkflowRuntime.execute, and equivalent behavior in file_activity for direct exports. Keep existing textContent/content/binaryContent/fileInfo keys compatible. Current reads materialize whole contents; changing the output to a stream changes the contract and cannot be a silent replacement.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of read. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Write File: source map

Trace

Saved identity: type=file; config.operation=write. Write File is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation write	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation write	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Write File, select type=file and operation=write, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of write. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



List Files: source map

Trace

Saved identity: type=file; config.operation=list. List Files is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation list	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation list	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing List Files, select type=file and operation=list, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of list. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Delete File: source map

Trace

Saved identity: type=file; config.operation=delete. Delete File is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation delete	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation delete	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Delete File, select type=file and operation=delete, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of delete. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and execute it in a clean test environment.

Do not patch only an already-generated archive.



Rename File: source map

Trace

Saved identity: type=file; config.operation=rename. Rename File is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation rename	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation rename	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Rename File, select type=file and operation=rename, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of rename. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Copy File: source map

Developer guide / Activities / File

Trace

Saved identity: type=file; config.operation=copy. Copy File is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation copy	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation copy	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Copy File, select type=file and operation=copy, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of copy. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



File Poller: source map

Trace

Saved identity: type=file; config.operation=poll. File Poller is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'file':
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: file; select operation poll	frontend/src/ActivityEditor.tsx:420 if (n.type === "file")
Local file execution; read is the fall-through after write/list/delete/rename/copy/poll branches	backend/app/runtime.py:1047 if activity.type == 'file':
Direct Python implementation; inspect operation poll	backend/app/raw_python_support/activities.py:50 def file_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing File Poller, select type=file and operation=poll, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of poll. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



FTP Get: source map

Trace

Saved identity: type=ftp; config.operation=get. FTP Get is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def ftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
FTP operations using the selected operation	backend/app/runtime.py:1799 def ftp(
Direct Python implementation; inspect operation get	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing FTP Get, select type=ftp and operation=get, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



FTP Put: source map

Trace

Saved identity: type=ftp; config.operation=put. FTP Put is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def ftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
FTP operations using the selected operation	backend/app/runtime.py:1799 def ftp(
Direct Python implementation; inspect operation put	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing FTP Put, select type=ftp and operation=put, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of put. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



FTP Delete: source map

Trace

Saved identity: type=ftp; config.operation=delete. FTP Delete is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def ftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
FTP operations using the selected operation	backend/app/runtime.py:1799 def ftp(
Direct Python implementation; inspect operation delete	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing FTP Delete, select type=ftp and operation=delete, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of delete. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



FTP Directory: source map

Trace

Saved identity: type=ftp; config.operation=dir. FTP Directory is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def ftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
FTP operations using the selected operation	backend/app/runtime.py:1799 def ftp(
Direct Python implementation; inspect operation dir	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing FTP Directory, select type=ftp and operation=dir, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of dir. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



FTP Change Directory: source map

Trace

Saved identity: type=ftp; config.operation=change_dir. FTP Change Directory is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def ftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
FTP operations using the selected operation	backend/app/runtime.py:1799 def ftp(
Direct Python implementation; inspect operation change_dir	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing FTP Change Directory, select type=ftp and operation=change_dir, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of change_dir. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SFTP Get: source map

Trace

Saved identity: type=sftp; config.operation=get. SFTP Get is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def sftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
SFTP operations using the selected operation	backend/app/runtime.py:1818 def sftp(
Direct Python implementation; inspect operation get	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SFTP Get, select type=sftp and operation=get, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

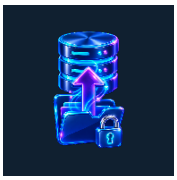
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SFTP Put: source map

Trace

Saved identity: type=sftp; config.operation=put. SFTP Put is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def sftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
SFTP operations using the selected operation	backend/app/runtime.py:1818 def sftp(
Direct Python implementation; inspect operation put	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SFTP Put, select type=sftp and operation=put, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

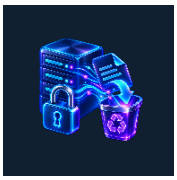
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of put. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SFTP Delete: source map

Trace

Saved identity: type=sftp; config.operation=delete. SFTP Delete is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def sftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
SFTP operations using the selected operation	backend/app/runtime.py:1818 def sftp(
Direct Python implementation; inspect operation delete	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SFTP Delete, select type=sftp and operation=delete, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of delete. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SFTP Directory: source map

Trace

Saved identity: type=sftp; config.operation=dir. SFTP Directory is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def sftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
SFTP operations using the selected operation	backend/app/runtime.py:1818 def sftp(
Direct Python implementation; inspect operation dir	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SFTP Directory, select type=sftp and operation=dir, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

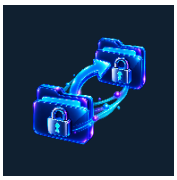
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of dir. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SFTP Change Directory: source map

Trace

Saved identity: type=sftp; config.operation=change_dir. SFTP Change Directory is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def sftp(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
SFTP operations using the selected operation	backend/app/runtime.py:1818 def sftp(
Direct Python implementation; inspect operation change_dir	backend/app/raw_python_support/activities.py:164 def transfer(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SFTP Change Directory, select type=sftp and operation=change_dir, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of change_dir. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

HTTP Listener: source map

Trace

Saved identity: type=http_listener; config.operation=listen. HTTP Listener is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (http_listener)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: http_listener; select operation listen	frontend/src/ActivityEditor.tsx:531 if (n.type === "http_listener")
Engine http_listener dispatch; operation listen	backend/app/runtime.py:750 async def execute(self,
Direct helper dispatch for http_listener; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing HTTP Listener, select type=http_listener and operation=listen, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of listen. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



HTTP Request: source map

Trace

Saved identity: type=http; config.operation=request. HTTP Request is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
HTTP client request	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation request	backend/app/raw_python_support/activities.py:199 def http_request(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing HTTP Request, select type=http and operation=request, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of request. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Read Server-Sent Events: source map

Trace

Saved identity: type=http; config.operation=sse_read. Read Server-Sent Events is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
HTTP client request	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation sse_read	backend/app/raw_python_support/activities.py:199 def http_request(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Read Server-Sent Events, select type=http and operation=sse_read, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of sse_read. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

WebSocket Exchange: source map

Trace

Saved identity: type=http; config.operation=websocket_exchange. WebSocket Exchange is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
HTTP client request	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation websocket_exchange	backend/app/raw_python_support/activities.py:199 def http_request(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing WebSocket Exchange, select type=http and operation=websocket_exchange, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of websocket_exchange. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

HTTP Write Preconditions: source map

Trace

Saved identity: type=http; config.operation=write_precondition. HTTP Write Preconditions is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
HTTP client request	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation write_precondition	backend/app/raw_python_support/activities.py:199 def http_request(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing HTTP Write Preconditions, select type=http and operation=write_precondition, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of write_precondition. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

HTTP Response: source map

Trace

Saved identity: type=http_response; config.operation=response. HTTP Response is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (http_response)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: http_response; select operation response	frontend/src/ActivityEditor.tsx:617 if (n.type === "http_response")
Engine http_response dispatch; operation response	backend/app/runtime.py:871 if activity.type == 'http_response':
Direct helper dispatch for http_response; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing HTTP Response, select type=http_response and operation=response, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of response. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

REST API Receiver: source map

Trace

Saved identity: type=rest; config.operation=receiver. REST API Receiver is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
REST outbound request; inbound path enters invoke_listener	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation receiver	backend/app/raw_python_support/activities.py:199 def http_request(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing REST API Receiver, select type=rest and operation=receiver, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of receiver. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



REST API Invoke: source map

Trace

Saved identity: type=rest; config.operation=invoke. REST API Invoke is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'http' or
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
REST outbound request; inbound path enters invoke_listener	backend/app/runtime.py:1034 if activity.type == 'http' or
Direct Python implementation; inspect operation invoke	backend/app/raw_python_support/activities.py:199 def http_request(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: http_capabilities	backend/tests/test_http_capabilities.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing REST API Invoke, select type=rest and operation=invoke, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of invoke. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_http_capabilities tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SOAP Service: source map

Trace

Saved identity: type=soap; config.operation=service. SOAP Service is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'soap' and
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: soap; select operation service	frontend/src/ActivityEditor.tsx:704 if (n.type === "soap")
SOAP outbound request; service path enters invoke_listener	backend/app/runtime.py:1040 if activity.type == 'soap' and
Direct Python implementation; inspect operation service	backend/app/raw_python_support/activities.py:199 def http_request(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SOAP Service, select type=soap and operation=service, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of service. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



SOAP Request Reply: source map

Trace

Saved identity: type=soap; config.operation=request_reply. SOAP Request Reply is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: if activity.type == 'soap' and
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: soap; select operation request_reply	frontend/src/ActivityEditor.tsx:704 if (n.type === "soap")
SOAP outbound request; service path enters invoke_listener	backend/app/runtime.py:1040 if activity.type == 'soap' and
Direct Python implementation; inspect operation request_reply	backend/app/raw_python_support/activities.py:199 def http_request(
Inbound HTTP/REST/SOAP hosting and response boundary	backend/app/main.py:2140 async def invoke_listener(
Direct-export application hosting	backend/app/raw_python_support/main.py:24 async def run_application(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing SOAP Request Reply, select type=soap and operation=request_reply, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

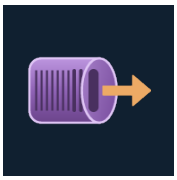
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of request_reply. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Queue Receiver: source map

Trace

Saved identity: type=ems; config.operation=queue_receiver. EMS Queue Receiver is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation queue_receiver	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Queue Receiver, select type=ems and operation=queue_receiver, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

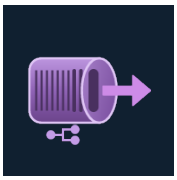
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of queue_receiver. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Topic Subscriber: source map

Trace

Saved identity: type=ems; config.operation=topic_subscriber. EMS Topic Subscriber is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation topic_subscriber	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Topic Subscriber, select type=ems and operation=topic_subscriber, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

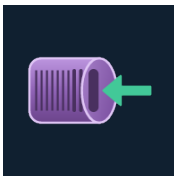
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of topic_subscriber. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Queue Sender: source map

Trace

Saved identity: type=ems; config.operation=send. EMS Queue Sender is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation send	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Queue Sender, select type=ems and operation=send, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

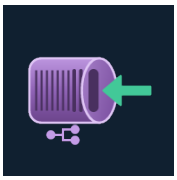
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of send. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Topic Publisher: source map

Trace

Saved identity: type=ems; config.operation=publish. EMS Topic Publisher is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation publish	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Topic Publisher, select type=ems and operation=publish, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

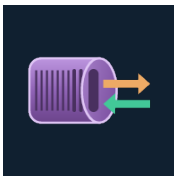
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of publish. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Request Reply: source map

Trace

Saved identity: type=ems; config.operation=request_reply. EMS Request Reply is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation request_reply	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Request Reply, select type=ems and operation=request_reply, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

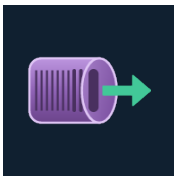
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of request_reply. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



EMS Reply: source map

Trace

Saved identity: type=ems; config.operation=reply. EMS Reply is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
EMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation reply	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing EMS Reply, select type=ems and operation=reply, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

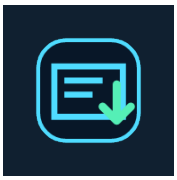
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of reply. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Get JMS Queue Message: source map

Trace

Saved identity: type=jms; config.operation=get_queue_message. Get JMS Queue Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation get_queue_message	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Get JMS Queue Message, select type=jms and operation=get_queue_message, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get_queue_message. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



JMS Receive Message: source map

Trace

Saved identity: type=jms; config.operation=receive_message. JMS Receive Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation receive_message	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JMS Receive Message, select type=jms and operation=receive_message, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of receive_message. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

JMS Request Reply: source map

Trace

Saved identity: type=jms; config.operation=request_reply. JMS Request Reply is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation request_reply	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JMS Request Reply, select type=jms and operation=request_reply, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of request_reply. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



JMS Send Message: source map

Trace

Saved identity: type=jms; config.operation=send_message. JMS Send Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation send_message	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JMS Send Message, select type=jms and operation=send_message, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of send_message. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Reply to JMS Message: source map

Trace

Saved identity: type=jms; config.operation=reply_message. Reply to JMS Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation reply_message	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Reply to JMS Message, select type=jms and operation=reply_message, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of reply_message. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Wait for JMS Request: source map

Trace

Saved identity: type=jms; config.operation=wait_request. Wait for JMS Request is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JMS operation routing and body/ack handling	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation wait_request	backend/app/raw_python_support/connectors.py:44 async def jms(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: jms_publish_performance	backend/tests/test_jms_publish_performance.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Wait for JMS Request, select type=jms and operation=wait_request, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of wait_request. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_jms_publish_performance tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Kafka Receive Message: source map

Trace

Saved identity: type=kafka; config.operation=receive. Kafka Receive Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: kafka; select operation receive	frontend/src/ActivityEditor.tsx:994 if (n.type === "kafka")
Kafka producer/consumer operation dispatch	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation receive	backend/app/raw_python_support/connectors.py:228 async def kafka(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: publisher_reuse	backend/tests/test_publisher_reuse.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Kafka Receive Message, select type=kafka and operation=receive, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of receive. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_publisher_reuse tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Kafka Send Message: source map

Trace

Saved identity: type=kafka; config.operation=send. Kafka Send Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: kafka; select operation send	frontend/src/ActivityEditor.tsx:994 if (n.type === "kafka")
Kafka producer/consumer operation dispatch	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation send	backend/app/raw_python_support/connectors.py:228 async def kafka(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: publisher_reuse	backend/tests/test_publisher_reuse.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Kafka Send Message, select type=kafka and operation=send, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of send. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_publisher_reuse tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Kafka Get Messages: source map

Trace

Saved identity: type=kafka; config.operation=get. Kafka Get Messages is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: kafka; select operation get	frontend/src/ActivityEditor.tsx:994 if (n.type === "kafka")
Kafka producer/consumer operation dispatch	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation get	backend/app/raw_python_support/connectors.py:228 async def kafka(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: publisher_reuse	backend/tests/test_publisher_reuse.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Kafka Get Messages, select type=kafka and operation=get, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_publisher_reuse tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Pub/Sub Subscriber: source map

Trace

Saved identity: type=pubsub; config.operation=subscribe. Pub/Sub Subscriber is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: pubsub; select operation subscribe	frontend/src/ActivityEditor.tsx:1021 if (n.type === "pubsub")
Pub/Sub publish/receive dispatch	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation subscribe	backend/app/raw_python_support/connectors.py:317 async def pubsub(
Credential/channel client factory	backend/app/google_pubsub.py:186 def create_client(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: google_pubsub	backend/tests/test_google_pubsub.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Pub/Sub Subscriber, select type=pubsub and operation=subscribe, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of subscribe. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_google_pubsub tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Pub/Sub Publisher: source map

Trace

Saved identity: type=pubsub; config.operation=publish. Pub/Sub Publisher is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def messaging(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: pubsub; select operation publish	frontend/src/ActivityEditor.tsx:1021 if (n.type === "pubsub")
Pub/Sub publish/receive dispatch	backend/app/runtime.py:1185 async def messaging(
Direct Python implementation; inspect operation publish	backend/app/raw_python_support/connectors.py:317 async def pubsub(
Credential/channel client factory	backend/app/google_pubsub.py:186 def create_client(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: google_pubsub	backend/tests/test_google_pubsub.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Pub/Sub Publisher, select type=pubsub and operation=publish, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of publish. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_google_pubsub tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Dynamic Connection: source map

Developer guide / Activities / SAP ECC

Trace

Saved identity: type=sap; config.operation=dynamic_connection. Dynamic Connection is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation dynamic_connection	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation dynamic_connection	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Dynamic Connection, select type=sap and operation=dynamic_connection, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

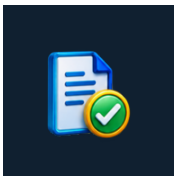
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of dynamic_connection. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Acknowledgment: source map

Trace

Saved identity: type=sap; config.operation=idoc_acknowledgment. IDoc Acknowledgment is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_acknowledgment	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_acknowledgment	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Acknowledgment, select type=sap and operation=idoc_acknowledgment, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_acknowledgment. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

IDoc Confirmation: source map

Trace

Saved identity: type=sap; config.operation=idoc_confirmation. IDoc Confirmation is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_confirmation	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_confirmation	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Confirmation, select type=sap and operation=idoc_confirmation, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_confirmation. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Converter: source map

Trace

Saved identity: type=sap; config.operation=idoc_converter. IDoc Converter is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_converter	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_converter	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Converter, select type=sap and operation=idoc_converter, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_converter. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Listener: source map

Trace

Saved identity: type=sap; config.operation=idoc_listener. IDoc Listener is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_listener	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_listener	backend/app/raw_python_support/connectors.py:197 async def sap(
Inbound SAP adapter	backend/app/sap.py:808 async def receive_idoc(
Continuous SAP jobs	backend/app/main.py:494 async def _continuous_sap_event_loop(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Listener, select type=sap and operation=idoc_listener, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

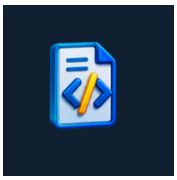
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_listener. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Parser: source map

Trace

Saved identity: type=sap; config.operation=idoc_parser. IDoc Parser is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_parser	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_parser	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Parser, select type=sap and operation=idoc_parser, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_parser. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Reader: source map

Trace

Saved identity: type=sap; config.operation=idoc_reader. IDoc Reader is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_reader	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_reader	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Reader, select type=sap and operation=idoc_reader, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_reader. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Post IDoc to SAP: source map

Trace

Saved identity: type=sap; config.operation=post_idoc. Post IDoc to SAP is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation post_idoc	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation post_idoc	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Post IDoc to SAP, select type=sap and operation=post_idoc, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

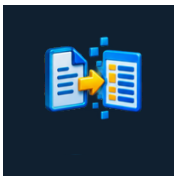
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of post_idoc. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



IDoc Renderer: source map

Trace

Saved identity: type=sap; config.operation=idoc_renderer. IDoc Renderer is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation idoc_renderer	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation idoc_renderer	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing IDoc Renderer, select type=sap and operation=idoc_renderer, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of idoc_renderer. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



RFC BAPI Listener: source map

Trace

Saved identity: type=sap; config.operation=rfc_bapi_listener. RFC BAPI Listener is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation rfc_bapi_listener	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation rfc_bapi_listener	backend/app/raw_python_support/connectors.py:197 async def sap(
Inbound SAP adapter	backend/app/sap.py:808 async def receive_idoc(
Continuous SAP jobs	backend/app/main.py:494 async def _continuous_sap_event_loop(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing RFC BAPI Listener, select type=sap and operation=rfc_bapi_listener, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

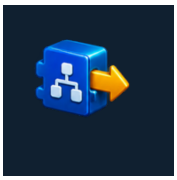
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of rfc_bapi_listener. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Invoke RFC BAPI in SAP: source map

Developer guide / Activities / SAP ECC

Trace

Saved identity: type=sap; config.operation=invoke_rfc_bapi. Invoke RFC BAPI in SAP is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation invoke_rfc_bapi	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation invoke_rfc_bapi	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Invoke RFC BAPI in SAP, select type=sap and operation=invoke_rfc_bapi, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

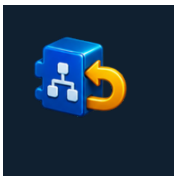
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of invoke_rfc_bapi. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Reply from RFC BAPI in SAP: source map

Trace

Saved identity: type=sap; config.operation=reply_rfc_bapi. Reply from RFC BAPI in SAP is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation reply_rfc_bapi	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation reply_rfc_bapi	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Reply from RFC BAPI in SAP, select type=sap and operation=reply_rfc_bapi, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

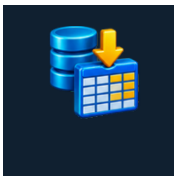
For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of reply_rfc_bapi. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Read Table: source map

Trace

Saved identity: type=sap; config.operation=read_table. Read Table is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/sap.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: sap; select operation read_table	frontend/src/ActivityEditor.tsx:1031 if (n.type === "sap")
SAP operation dispatch; listeners use receive_idoc	backend/app/sap.py:1096 def execute(
Direct Python implementation; inspect operation read_table	backend/app/raw_python_support/connectors.py:197 async def sap(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: sap_mapper	backend/tests/test_sap_mapper.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Read Table, select type=sap and operation=read_table, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of read_table. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_sap_mapper tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



JDBC Create: source map

Trace

Saved identity: type=jdbc; config.operation=insert. JDBC Create is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation insert	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation insert	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JDBC Create, select type=jdbc and operation=insert, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of insert. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



JDBC Update: source map

Trace

Saved identity: type=jdbc; config.operation=update. JDBC Update is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation update	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation update	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JDBC Update, select type=jdbc and operation=update, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of update. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



JDBC Query: source map

Trace

Saved identity: type=jdbc; config.operation=query. JDBC Query is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation query	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation query	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JDBC Query, select type=jdbc and operation=query, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of query. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



JDBC Truncate: source map

Trace

Saved identity: type=jdbc; config.operation=truncate. JDBC Truncate is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation truncate	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation truncate	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JDBC Truncate, select type=jdbc and operation=truncate, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of truncate. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



JDBC Delete: source map

Trace

Saved identity: type=jdbc; config.operation=delete. JDBC Delete is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation delete	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation delete	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing JDBC Delete, select type=jdbc and operation=delete, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of delete. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Stored Procedure: source map

Trace

Saved identity: type=jdbc; config.operation=call. Stored Procedure is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation call	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation call	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Stored Procedure, select type=jdbc and operation=call, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of call. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Dynamic SQL: source map

Trace

Saved identity: type=jdbc; config.operation=dynamic. Dynamic SQL is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/jdbc.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: jdbc; select operation dynamic	frontend/src/ActivityEditor.tsx:766 if (n.type === "jdbc")
JDBC adapter; uses transaction when supplied	backend/app/jdbc.py:336 def execute(
Direct Python implementation; inspect operation dynamic	backend/app/raw_python_support/connectors.py:154 async def jdbc(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Dynamic SQL, select type=jdbc and operation=dynamic, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of dynamic. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Snowflake Insert: source map

Trace

Saved identity: type=snowflake; config.operation=insert. Snowflake Insert is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/snowflake.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: snowflake; select operation insert	frontend/src/ActivityEditor.tsx:800 if (n.type === "snowflake")
Snowflake operation adapter	backend/app/snowflake.py:240 def execute(
Direct Python implementation; inspect operation insert	backend/app/raw_python_support/connectors.py:159 async def snowflake(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: snowflake	backend/tests/test_snowflake.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Snowflake Insert, select type=snowflake and operation=insert, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of insert. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_snowflake tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Snowflake Query: source map

Trace

Saved identity: type=snowflake; config.operation=query. Snowflake Query is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/snowflake.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: snowflake; select operation query	frontend/src/ActivityEditor.tsx:800 if (n.type === "snowflake")
Snowflake operation adapter	backend/app/snowflake.py:240 def execute(
Direct Python implementation; inspect operation query	backend/app/raw_python_support/connectors.py:159 async def snowflake(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: snowflake	backend/tests/test_snowflake.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Snowflake Query, select type=snowflake and operation=query, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of query. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_snowflake tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Snowflake Update: source map

Trace

Saved identity: type=snowflake; config.operation=update. Snowflake Update is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/snowflake.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: snowflake; select operation update	frontend/src/ActivityEditor.tsx:800 if (n.type === "snowflake")
Snowflake operation adapter	backend/app/snowflake.py:240 def execute(
Direct Python implementation; inspect operation update	backend/app/raw_python_support/connectors.py:159 async def snowflake(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: snowflake	backend/tests/test_snowflake.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Snowflake Update, select type=snowflake and operation=update, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of update. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_snowflake tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Snowflake Delete: source map

Trace

Saved identity: type=snowflake; config.operation=delete. Snowflake Delete is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/snowflake.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: snowflake; select operation delete	frontend/src/ActivityEditor.tsx:800 if (n.type === "snowflake")
Snowflake operation adapter	backend/app/snowflake.py:240 def execute(
Direct Python implementation; inspect operation delete	backend/app/raw_python_support/connectors.py:159 async def snowflake(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: snowflake	backend/tests/test_snowflake.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Snowflake Delete, select type=snowflake and operation=delete, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of delete. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_snowflake tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Snowflake Bulk Load: source map

Trace

Saved identity: type=snowflake; config.operation=bulk_load. Snowflake Bulk Load is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/snowflake.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: snowflake; select operation bulk_load	frontend/src/ActivityEditor.tsx:800 if (n.type === "snowflake")
Snowflake operation adapter	backend/app/snowflake.py:240 def execute(
Direct Python implementation; inspect operation bulk_load	backend/app/raw_python_support/connectors.py:159 async def snowflake(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: snowflake	backend/tests/test_snowflake.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Snowflake Bulk Load, select type=snowflake and operation=bulk_load, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of bulk_load. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_snowflake tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



AMQP Receive Message: source map

Trace

Saved identity: type=amqp; config.operation=receive. AMQP Receive Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/amqp.py :: def send(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: amqp; select operation receive	frontend/src/ActivityEditor.tsx:791 if (n.type === "amqp")
AMQP send adapter; get handles receives	backend/app/amqp.py:135 def send(
Direct Python implementation; inspect operation receive	backend/app/raw_python_support/connectors.py:164 async def amqp(
Receive adapter and acknowledgement callback	backend/app/amqp.py:182 def get(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing AMQP Receive Message, select type=amqp and operation=receive, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of receive. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



AMQP Get Message: source map

Trace

Saved identity: type=amqp; config.operation=get. AMQP Get Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/amqp.py :: def send(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: amqp; select operation get	frontend/src/ActivityEditor.tsx:791 if (n.type === "amqp")
AMQP send adapter; get handles receives	backend/app/amqp.py:135 def send(
Direct Python implementation; inspect operation get	backend/app/raw_python_support/connectors.py:164 async def amqp(
Receive adapter and acknowledgement callback	backend/app/amqp.py:182 def get(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing AMQP Get Message, select type=amqp and operation=get, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



AMQP Send Message: source map

Trace

Saved identity: type=amqp; config.operation=send. AMQP Send Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/amqp.py :: def send(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: amqp; select operation send	frontend/src/ActivityEditor.tsx:791 if (n.type === "amqp")
AMQP send adapter; get handles receives	backend/app/amqp.py:135 def send(
Direct Python implementation; inspect operation send	backend/app/raw_python_support/connectors.py:164 async def amqp(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing AMQP Send Message, select type=amqp and operation=send, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of send. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



AMQP Dead Letter Message: source map

Trace

Saved identity: type=amqp; config.operation=dead_letter. AMQP Dead Letter Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/amqp.py :: def send(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: amqp; select operation dead_letter	frontend/src/ActivityEditor.tsx:791 if (n.type === "amqp")
AMQP send adapter; get handles receives	backend/app/amqp.py:135 def send(
Direct Python implementation; inspect operation dead_letter	backend/app/raw_python_support/connectors.py:164 async def amqp(
Receive adapter and acknowledgement callback	backend/app/amqp.py:182 def get(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing AMQP Dead Letter Message, select type=amqp and operation=dead_letter, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of dead_letter. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Parse XML: source map

Trace

Saved identity: type=xml; config.operation=parse. Parse XML is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def parse_xml_activity(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
XML parse; render_xml_activity handles render	backend/app/data_formats.py:351 def parse_xml_activity(
Direct Python implementation; inspect operation parse	backend/app/raw_python_support/activities.py:134 def data_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Parse XML, select type=xml and operation=parse, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of parse. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Render XML: source map

Trace

Saved identity: type=xml; config.operation=render. Render XML is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def parse_xml_activity(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
XML parse; render_xml_activity handles render	backend/app/data_formats.py:351 def parse_xml_activity(
Direct Python implementation; inspect operation render	backend/app/raw_python_support/activities.py:134 def data_activity(
XML render implementation	backend/app/data_formats.py:382 def render_xml_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Render XML, select type=xml and operation=render, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of render. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Parse JSON: source map

Trace

Saved identity: type=json; config.operation=parse. Parse JSON is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def json_activity(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JSON parse/render and validation	backend/app/data_formats.py:512 def json_activity(
Direct Python implementation; inspect operation parse	backend/app/raw_python_support/activities.py:134 def data_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Parse JSON, select type=json and operation=parse, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of parse. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Render JSON: source map

Trace

Saved identity: type=json; config.operation=render. Render JSON is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def json_activity(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
JSON parse/render and validation	backend/app/data_formats.py:512 def json_activity(
Direct Python implementation; inspect operation render	backend/app/raw_python_support/activities.py:134 def data_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Render JSON, select type=json and operation=render, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of render. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Parse Data: source map

Trace

Saved identity: type=flat; config.operation=parse. Parse Data is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def flat_data(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Flat record parse/render	backend/app/data_formats.py:677 def flat_data(
Direct Python implementation; inspect operation parse	backend/app/raw_python_support/activities.py:134 def data_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Parse Data, select type=flat and operation=parse, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of parse. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.



Render Data: source map

Trace

Saved identity: type=flat; config.operation=render. Render Data is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/data_formats.py :: def flat_data(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Flat record parse/render	backend/app/data_formats.py:677 def flat_data(
Direct Python implementation; inspect operation render	backend/app/raw_python_support/activities.py:134 def data_activity(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: activity_packs	backend/tests/test_activity_packs.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Render Data, select type=flat and operation=render, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of render. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_activity_packs tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Read Excel Workbook: source map

Trace

Saved identity: type=excel; config.operation=read. Read Excel Workbook is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: def read_excel(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: excel; select operation read	frontend/src/ActivityEditor.tsx:842 if (n.type === "excel")
Workbook reading	backend/app/runtime.py:1746 def read_excel(
Direct Python implementation; inspect operation read	backend/app/raw_python_support/activities.py:139 def excel_read(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: amqp_jdbc_excel_command	backend/tests/test_amqp_jdbc_excel_command.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Read Excel Workbook, select type=excel and operation=read, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

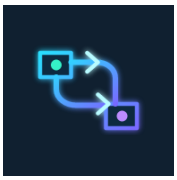
Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of read. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_amqp_jdbc_excel_command tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Mapper: source map

Trace

Saved identity: type=mapper; config.operation=map. Mapper is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/mapper.py :: def execute(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Schema mapping engine	backend/app/mapper.py:950 def execute(
Direct Python implementation; inspect operation map	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Mapper, select type=mapper and operation=map, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of map. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Transform: source map

Trace

Saved identity: type=dataweave; config.operation=transform. Transform is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/dataweave.py :: def execute_details(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: dataweave; select operation transform	frontend/src/ActivityEditor.tsx:1295 if (n.type === "dataweave")
Transformation script parser and evaluator	backend/app/dataweave.py:346 def execute_details(
Direct Python implementation; inspect operation transform	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Transform, select type=dataweave and operation=transform, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of transform. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Log: source map

Trace

Saved identity: type=log; config.operation=write. Log is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (log)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: log; select operation write	frontend/src/ActivityEditor.tsx:1306 if (n.type === "log")
Engine log dispatch; operation write	backend/app/runtime.py:876 if activity.type == 'log':
Direct helper dispatch for log; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Log, select type=log and operation=write, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of write. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Catch Exception: source map

Trace

Saved identity: type=catch; config.operation=catch. Catch Exception is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (catch)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: catch; select operation catch	frontend/src/ActivityEditor.tsx:401 if (n.type === "catch")
Engine catch dispatch; operation catch	backend/app/runtime.py:856 if activity.type == 'catch':
Direct helper dispatch for catch; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Catch Exception, select type=catch and operation=catch, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of catch. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Throw Exception: source map

Trace

Saved identity: type=throw; config.operation=throw. Throw Exception is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (throw)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: throw; select operation throw	frontend/src/ActivityEditor.tsx:408 if (n.type === "throw")
Engine throw dispatch; operation throw	backend/app/runtime.py:857 if activity.type == 'throw':
Direct helper dispatch for throw; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Throw Exception, select type=throw and operation=throw, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of throw. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Rethrow Exception: source map

Trace

Saved identity: type=rethrow; config.operation=rethrow. Rethrow Exception is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (rethrow)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: rethrow; select operation rethrow	frontend/src/ActivityEditor.tsx:415 if (n.type === "rethrow")
Engine rethrow dispatch; operation rethrow	backend/app/runtime.py:865 if activity.type == 'rethrow':
Direct helper dispatch for rethrow; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Rethrow Exception, select type=rethrow and operation=rethrow, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of rethrow. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Confirm Message: source map

Trace

Saved identity: type=confirm; config.operation=acknowledge. Confirm Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (confirm)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: confirm; select operation acknowledge	frontend/src/ActivityEditor.tsx:387 if (n.type === "confirm")
Engine confirm dispatch; operation acknowledge	backend/app/runtime.py:909 if activity.type == 'confirm':
Direct helper dispatch for confirm; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Confirm Message, select type=confirm and operation=acknowledge, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of acknowledge. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_tasks_runtime tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Java Activity: source map

Trace

Saved identity: type=java; config.operation=invoke. Java Activity is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def java_worker(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: java; select operation invoke	frontend/src/ActivityEditor.tsx:1328 if (n.type === "java")
Java invocation worker	backend/app/runtime.py:1834 async def java_worker(
Direct Python implementation; inspect operation invoke	backend/app/raw_python_support/activities.py:249 async def java_invoke(
Python/native worker lifecycle	backend/app/java_bridge.py:1 (module)
Native Java connector implementation	java-bridge/src/com/mina/bridge/MinaJavaBridge.java:1 (module)
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Java Activity, select type=java and operation=invoke, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of invoke. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment. Do not patch only an already-generated archive.

Python Invoke: source map

Developer guide / Activities / General

Trace

Saved identity: type=python; config.operation=invoke. Python Invoke is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: async def python_worker(
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: python; select operation invoke	frontend/src/ActivityEditor.tsx:1356 if (n.type === "python")
Python invocation worker	backend/app/runtime.py:1889 async def python_worker(
Direct Python implementation; inspect operation invoke	backend/app/raw_python_support/activities.py:225 async def python_invoke(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Python Invoke, select type=python and operation=invoke, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of invoke. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

External Command: source map

Trace

Saved identity: type=basic; config.operation=external_command. External Command is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation external_command	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation external_command	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing External Command, select type=basic and operation=external_command, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of external_command. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Assign: source map

Trace

Saved identity: type=basic; config.operation=assign. Assign is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation assign	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation assign	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Assign, select type=basic and operation=assign, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of assign. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Checkpoint: source map

Trace

Saved identity: type=basic; config.operation=checkpoint. Checkpoint is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation checkpoint	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation checkpoint	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Checkpoint, select type=basic and operation=checkpoint, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of checkpoint. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Deduplicate Message: source map

Trace

Saved identity: type=basic; config.operation=deduplicate. Deduplicate Message is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation deduplicate	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation deduplicate	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Deduplicate Message, select type=basic and operation=deduplicate, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of deduplicate. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Message Receipt: source map

Trace

Saved identity: type=basic; config.operation=message_receipt. Message Receipt is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation message_receipt	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation message_receipt	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Message Receipt, select type=basic and operation=message_receipt, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of message_receipt. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Replay Messages: source map

Developer guide / Activities / General

Trace

Saved identity: type=basic; config.operation=replay_messages. Replay Messages is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation replay_messages	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation replay_messages	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Replay Messages, select type=basic and operation=replay_messages, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of replay_messages. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Aggregate Messages: source map

Trace

Saved identity: type=basic; config.operation=aggregate_messages. Aggregate Messages is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation aggregate_messages	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation aggregate_messages	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Aggregate Messages, select type=basic and operation=aggregate_messages, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of aggregate_messages. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Split Records: source map

Developer guide / Activities / General

Trace

Saved identity: type=basic; config.operation=split_records. Split Records is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation split_records	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation split_records	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Split Records, select type=basic and operation=split_records, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of split_records. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Sleep: source map

Developer guide / Activities / General

Trace

Saved identity: type=basic; config.operation=sleep. Sleep is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation sleep	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation sleep	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Sleep, select type=basic and operation=sleep, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of sleep. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.



Get Shared Variable: source map

Trace

Saved identity: type=basic; config.operation=get_shared_variable. Get Shared Variable is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation get_shared_variable	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation get_shared_variable	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Get Shared Variable, select type=basic and operation=get_shared_variable, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of get_shared_variable. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Set Shared Variable: source map

Trace

Saved identity: type=basic; config.operation=set_shared_variable. Set Shared Variable is a palette definition rendered by shared React components, not necessarily its own frontend file or Python module.

```
main.tsx packs -> addActivity -> ActivityEditor
Project.tasks[].activities[] -> Run API / DebugManager
WorkflowRuntime.execute_with_policy -> execute
backend/app/runtime.py :: execute (basic)
record_activity_output -> downstream mappings / Job Data
```

For event starters the API/listener loop supplies event_output before the graph continues; a manual single-activity test is not equivalent to the continuous listener lifecycle.

Source locations

Responsibility	Source location / search anchor
Palette labels, types, operations and icons	frontend/src/main.tsx:359 const packs:
Create a canvas activity and its initial config	frontend/src/main.tsx:2089 const addActivity =
Configuration/Input/Output/Errors contract	frontend/src/ActivityEditor.tsx:310 export function activityContract(
Project-specific schema resolution	frontend/src/ActivityEditor.tsx:1435 function resolvedActivityContract(
Configuration keys exposed for mapping	frontend/src/ActivityEditor.tsx:1564 function runtimeMappableInputs(
Shared configuration field renderer	frontend/src/ActivityEditor.tsx:2328 function FieldEditor(
Input mappings editor	frontend/src/ActivityEditor.tsx:2769 function InputEditor(
Output schema editor	frontend/src/ActivityEditor.tsx:3541 function OutputEditor(
Advanced logging/retry UI	frontend/src/ActivityEditor.tsx:3591 function AdvancedEditor(
Error handling UI	frontend/src/ActivityEditor.tsx:3752 function ErrorEditor(
Persisted activity model	backend/app/models.py:21 class Activity(
Run HTTP API	backend/app/main.py:739 @app.post('/api/projects/{project_id}/run')
Graph scheduler	backend/app/runtime.py:375 async def run(self,
Retry/logging policy wrapper	backend/app/runtime.py:653 async def execute_with_policy(

Responsibility	Source location / search anchor
Resolve configuration and input mappings	backend/app/runtime.py:713 def resolve_activity_config(
Activity dispatch	backend/app/runtime.py:750 async def execute(self,
Publish outputs and captured inputs	backend/app/runtime.py:602 def record_activity_output(
Debug stepping over the runtime	backend/app/debugger.py:323 async def step(
Contract branch: basic; select operation set_shared_variable	frontend/src/ActivityEditor.tsx:1367 if (n.type === "basic")
Engine basic dispatch; operation set_shared_variable	backend/app/runtime.py:788 if activity.type == 'basic':
Direct helper dispatch for basic; task graph may also be generated inline	backend/app/raw_python_support/core.py:266 async def execute(
Direct graph/code generation	backend/app/raw_python.py:663 def raw_python_files(
Export capability selection	backend/app/raw_python.py:63 def _activity_capabilities(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

What to change

When changing Set Shared Variable, select type=basic and operation=set_shared_variable, not merely the displayed label. The shared dispatcher may serve several activities. Add/adjust editor fields, resolved input values and returned output keys together; preserve behavior of the other operations sharing this handler.

Configuration tab: activityContract and FieldEditor. Input tab: runtimeMappableInputs/InputEditor; mappings persist as config.inputMappings and are resolved before execute. Output tab: contract/resolved schema plus actual returned keys. Advanced tab: advancedDefaults/AdvancedEditor and execute_with_policy. Errors tab: declared faults/ErrorEditor plus real exceptions and run/debug propagation. UI declarations do not automatically implement runtime behavior.

For provider operations, inspect the adapter and Java bridge where listed. Moving work to a thread does not automatically make native I/O cancellable. Preserve acknowledgement, session ownership and worker cleanup when changing performance or timeouts.

Tests and exports

Relevant suites are starting points, not a claim of exhaustive coverage of set_shared_variable. Add a regression that executes this exact type/operation with resolved mappings and asserts the returned shape. Exercise empty input, invalid values, provider errors and cancellation as applicable.

```
cd backend
python -m unittest tests.test_mapper_code_general tests.test_raw_python tests.test_engine_export
```

Engine-based export reuses/prunes engine modules. Raw Python export generates task code and specializes raw_python_support; review both paths and capability filtering. Re-export a fixture and execute it in a clean test environment.

Do not patch only an already-generated archive.

Tasks, sub-tasks, persistence and copy

Developer guide / Features

Responsibility

Tasks own activity/transition/group graphs. Calls use saved task references; activity IDs also participate in mapping expressions.

```
Task editor -> normalizeProject -> persistProject -> API -> store.save_project
Call Sub Task -> runtime call_task branch -> child run -> caller output
```

Source locations

Responsibility	Source location / search anchor
Loaded-project normalization	frontend/src/main.tsx:1247 const normalizeProject =
Browser save API	frontend/src/main.tsx:2492 const persistProject =
Task cloning	frontend/src/copyTask.ts:2 export function copyTask
Legacy copy reference repair	frontend/src/repairCopiedTask.ts:1 (module)
Python task schema	backend/app/models.py:108 class ProcessDefinition(
Disk persistence	backend/app/store.py:67 def save_project(
Nested execution	backend/app/runtime.py:1160 if activity.type == 'call_task':
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)

Safe change procedure

For a new task setting, change both the frontend Task representation and ProcessDefinition validation/defaults. Preserve IDs during a copy or rewrite references consistently. Test nested calls, failed children, interfaces, old imports and missing referenced tasks. A renamed label must not silently break expressions.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Run, continuous events and Stop

Developer guide / Features

Responsibility

Run coordinates application listeners/jobs while WorkflowRuntime schedules activities and groups.

```
main.tsx run -> POST /projects/{id}/run -> listener registration or runtime.run
Stop -> stop_project -> cancel jobs -> connector cleanup
```

Source locations

Responsibility	Source location / search anchor
Run action	frontend/src/main.tsx:2673 const run = async
Run API	backend/app/main.py:740 async def run(project_id:
Broker/event polling lifecycle	backend/app/main.py:348 async def _continuous_event_loop(
SAP event lifecycle	backend/app/main.py:494 async def _continuous_sap_event_loop(
Stop endpoint	backend/app/main.py:791 async def stop_project(
Graph execution	backend/app/runtime.py:375 async def run(self,
Packaged engine command-line host	backend/run_deployment.py:1 (module)
Regression suite: tasks_runtime	backend/tests/test_tasks_runtime.py:1 (module)
Regression suite: publisher_reuse	backend/tests/test_publisher_reuse.py:1 (module)

Safe change procedure

Distinguish a one-shot run from a long-lived event starter. Preserve transport context and acknowledgement timing when spawning jobs. For Stop, test active I/O, idle listeners and worker cleanup. Do not equate cancelling an asyncio task with undoing an external send.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Debug controls, test bench and Job Data

Developer guide / Features

Responsibility

Debug maintains sessions, frames, breakpoints, watches, captured inputs/outputs and isolated activity tests.

```
DebugRibbon / DebugToolsDialog -> debugAction -> DebugManager.action
 -> step / runtime.execute_with_policy -> DebugManager.view -> Job Data
```

Source locations

Responsibility	Source location / search anchor
Start debug request	frontend/src/main.tsx:2713 debug = async
Debug action transport	frontend/src/main.tsx:2776 debugAction = async
Ribbon controls	frontend/src/main.tsx:6642 function DebugRibbon(
Test bench/tools	frontend/src/main.tsx:6548 function DebugToolsDialog(
Input/output/variables viewer	frontend/src/DebugJobData.tsx:1 (module)
Activity-selection tree	frontend/src/DebugActivityTree.tsx:1 (module)
Nested task traversal	frontend/src/debugTaskTree.ts:1 (module)
Test payload editor	frontend/src/DebugPayloadEditor.tsx:1 (module)
Debug API	backend/app/main.py:2041 async def start_debug(
Debug commands	backend/app/debugger.py:159 async def action(
Frame execution	backend/app/debugger.py:323 async def step(
Session cleanup	backend/app/debugger.py:296 async def stop(
Serializable debug state	backend/app/debugger.py:447 def view(
Resolved input/output capture	backend/app/runtime.py:602 def record_activity_output(
Regression suite: debug_testing	backend/tests/test_debug_testing.py:1 (module)
Regression suite: debug_input_capture	backend/tests/test_debug_input_capture.py:1 (module)

Safe change procedure

For a new debug action update the client request, DebugAction model and DebugManager.action validation together. Capture resolved activity input separately from previous output. Test pause/resume, nested frames, injected data, run-once side effects and stop during I/O. Test bench execution can modify real systems.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Input mapping, schema mapper and suggestions

Developer guide / Features

Responsibility

The generic input editor and dedicated schema mapper share scoped condition evaluation. Parentheses, logical precedence, negation and comparisons preserve quoted literals and references. The same evaluator is packaged for Python Direct transitions and control groups.

```
DataSourcePane -> MappingBinding / InputEditor -> config.inputMappings
resolve_activity_config -> map_input_values -> evaluate_mapping / resolve
Mapper activity -> mapper.execute -> output validation
```

Source locations

Responsibility	Source location / search anchor
Available upstream Data tree	frontend/src/ActivityEditor.tsx:2482 export function upstreamActivitySources(
Data/functions/properties browser	frontend/src/ActivityEditor.tsx:2535 export function DataSourcePane(
Target field control	frontend/src/ActivityEditor.tsx:2641 function MappingBinding(
Expanded mapping popup	frontend/src/ActivityEditor.tsx:2736 function ExpandedInputMappingDialog(
Dedicated schema mapper	frontend/src/MapperStudio.tsx:56 export default function MapperStudio
Completion UI/keyboard behavior	frontend/src/MappingExpressionInput.tsx:1 (module)
Suggestion matching	frontend/src/mappingCompletion.ts:1 export function mappingPathSuggestions
Input mapping evaluation	backend/app/runtime.py:1511 def map_input_values(
Structured rule evaluation	backend/app/runtime.py:1592 def evaluate_mapping(
Reference/expression resolution	backend/app/runtime.py:1664 def resolve(self,
Transformation mapping	backend/app/mapper.py:950 def execute(
Map & Test endpoint	backend/app/main.py:1839 def mapper_test(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: sap_idoc_arrays	backend/tests/test_sap_idoc_arrays.py:1 (module)

Safe change procedure

For a new expression syntax update completion/parser/runtime together. Preserve structured mapping rules and array cardinality; never turn expression strings into unrestricted eval. Compare Map & Test with Run, Debug and direct-export

behavior using the same fixture.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Built-in and project functions

Developer guide / Features

Responsibility

The function catalog controls discoverability; Python evaluation controls results. Project custom functions are available in Mapper, generic Input mappings, Map & Test and both Python archive formats, including nested calls and repeating-element scope. Calls enforce declared arity and a recursion limit.

```
mapper-functions.ts template -> expression mapping -> evaluate_function_expression
 -> mapper.apply_function / project custom-function expression
```

Source locations

Responsibility	Source location / search anchor
All builtin catalog entries	frontend/src/mapper-functions.ts:1 (module)
Function picker and custom function editing	frontend/src/ActivityEditor.tsx:2535 export function DataSourcePane(
Builtin implementation dispatch	backend/app/mapper.py:304 def apply_function(
Function expression evaluation	backend/app/runtime.py:1639 def evaluate_function_expression(
Persisted custom functions	backend/app/models.py:185 class Project(
Export function usage	backend/app/raw_python.py:663 def raw_python_files(
Regression suite: mapper_code_general	backend/tests/test_mapper_code_general.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Safe change procedure

Add the editor signature, description and insertion template; implement its Python semantics and input validation; update export dependencies if necessary. Test nulls, coercion, arrays, timezone/encoding and invalid arity. Maintain matching semantics in direct archives.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Canvas and mapper drag-and-drop

Developer guide / Features

Responsibility

Palette creation, moving canvas nodes, connecting transitions and mapping data are different gestures and data-transfer contracts.

```
Palette onDragStart -> canvas onDrop -> addActivity -> task.activities
Node pointer drag -> coordinates / selection -> saved position
DataSourcePane drag -> MappingBinding drop -> inputMappings
```

Source locations

Responsibility	Source location / search anchor
Palette drag source	frontend/src/main.tsx:3583 onDragStart={(e) =>
Canvas drop target	frontend/src/main.tsx:3647 onDrop={(e) =>
Activity creation	frontend/src/main.tsx:2089 const addActivity =
Editable-control shortcut guard	frontend/src/main.tsx:113 function editableControlFromEvent(
Transition geometry	frontend/src/main.tsx:196 function activityWireGeometry(
Picker insertion	frontend/src/ActivityPicker.tsx:16 export default function ActivityPicker
Mapping drag sources	frontend/src/ActivityEditor.tsx:2535 export function DataSourcePane(
Mapping drop targets	frontend/src/ActivityEditor.tsx:2641 function MappingBinding(
Group condition drop target	frontend/src/main.tsx:4232 function GroupExpressionTarget(
Selection visuals	frontend/src/canvas-selection.css:1 (module)

Safe change procedure

Do not change only the drop handler. Check source MIME/data payload, zoom/scroll coordinate conversion, event propagation, nested-group coordinates and undo/persistence. Preserve keyboard input by excluding editable controls from global shortcuts. Verify drag/drop in browser and packaged desktop, plus mapping into the expanded popup.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Groups: editor, graph boundaries and runtime state

Responsibility

A group includes visual membership plus scheduling, iteration variables, failure handling and resource lifecycle.

```
GroupEditor -> GroupDefinition -> compile_groups
enter_group_boundaries -> _begin_group -> activities
leave_group_boundaries / retry_failed_group -> _finish_group
```

Source locations

Responsibility	Source location / search anchor
Create group	frontend/src/main.tsx:2273 const addGroup =
Editor defaults	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group configuration	frontend/src/main.tsx:4252 function GroupEditor(
Group rendering	frontend/src/groups.css:1 (module)
Saved group schema	backend/app/models.py:28 class GroupDefinition(
Group graph plans	backend/app/runtime.py:162 def compile_groups(
Enter resources/conditions/iteration	backend/app/runtime.py:234 async def _begin_group(
Commit/rollback/release	backend/app/runtime.py:298 async def _finish_group(
Iteration and nested exit	backend/app/runtime.py:324 async def leave_group_boundaries(
Retry scope	backend/app/runtime.py:354 async def retry_failed_group(
Native transaction session	backend/app/jdbc.py:329 class JavaJdbcTransaction
Direct-export group planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Safe change procedure

Keep editor defaults, parent_group_id/member_activity_ids validation and runtime plans aligned. Test nested entry/exit, skip branches, retries, cancellation and restoration of loop variables. Transaction groups must close the exact owned connection on every exit. Critical sections are runtime-local, not distributed locks.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

If: source map

Developer guide / Groups

Behavior to preserve

Evaluates condition before entry; false skips the group.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
If defaults; match group type if	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

While True: source map

Developer guide / Groups

Behavior to preserve

Evaluates condition before each iteration, including the first.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
While True defaults; match group type while	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

For Each: source map

Behavior to preserve

Resolves the collection at entry and runs once per item. Range mode uses inclusive start/end with a nonzero increment.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
For Each defaults; match group type for-each	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Iterate: source map

Developer guide / Groups

Behavior to preserve

Resolves source once at entry and iterates its elements.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Iterate defaults; match group type iterate	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Repeat Until True: source map

Behavior to preserve

Runs once, checks condition at exit and repeats until true.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Repeat Until True defaults; match group type repeat	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Repeat on Error Until True: source map

Behavior to preserve

Retries the whole group after an unhandled fault while stopCondition is false. A true stop condition or exhausted retry count propagates the fault.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Repeat on Error Until True defaults; match group type repeat-on-error	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Scope: source map

Developer guide / Groups

Behavior to preserve

Runs the enclosed graph once. Nested groups retain their own semantics.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Scope defaults; match group type scope	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

None: source map

Behavior to preserve

Runs the enclosed graph once; this is an execution boundary, not a separate transaction.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
None defaults; match group type none	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Critical Section: source map

Developer guide / Groups

Behavior to preserve

Holds a runtime-local named lock until completion, failure or cancellation. This is not a distributed lock across machines or separate runtime processes.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Critical Section defaults; match group type critical-section	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

JDBC Transaction: source map

Behavior to preserve

Uses one database connection for matching JDBC operations, commits on successful exit and rolls back on an escaping fault or cancellation. External non-JDBC side effects are not rolled back.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
JDBC Transaction defaults; match group type transaction-jdbc	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Pick First: source map

Behavior to preserve

Chooses the first eligible transition in graph order. This is first-match routing, not concurrent event racing or fastest-response selection.

Set Group name, Parent group and direct activity membership. A member has one direct owner; use parent_group_id for nesting. Normal groups require a connected single-entry, single-exit graph. Pick First has specialized branch selection.

Use the group type branch in these shared methods. Do not create a UI-only group option; update validation, runtime entry/exit and direct-export scheduling together.

Source locations

Responsibility	Source location / search anchor
Pick First defaults; match group type pick-first	frontend/src/main.tsx:4246 function groupConfigDefaults(
Group-specific condition/iteration controls	frontend/src/main.tsx:4252 function GroupEditor(
Type-specific entry branch	backend/app/runtime.py:234 async def _begin_group(
Type-specific loop/exit branch	backend/app/runtime.py:324 async def leave_group_boundaries(
Error/retry handling	backend/app/runtime.py:354 async def retry_failed_group(
Resource release	backend/app/runtime.py:298 async def _finish_group(
Export nesting/graph planning	backend/app/raw_python.py:370 def _simple_group_plans(
Regression suite: groups_runtime	backend/tests/test_groups_runtime.py:1 (module)
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)

Regression focus

Test zero/one/multiple iterations or selected branches as applicable; nested groups; an activity fault; cancellation; and a saved/archive round trip. Verify variables, locks and sessions are restored or released at scope exit.

Engine archives versus direct Python generation

Responsibility

The two archive forms have different code-generation and packaging responsibilities.

```
PackageDialog -> package_project -> build_deployment_archive
engine_python_files -> pruned engine modules
raw_python_files -> task Python + specialized support modules
```

Source locations

Responsibility	Source location / search anchor
Packaging UI	frontend/src/main.tsx:4328 function PackageDialog(
Archive orchestration	backend/app/main.py:1379 def build_deployment_archive(
Capability registry	backend/app/raw_python.py:63 def _activity_capabilities(
Direct application generation	backend/app/raw_python.py:663 def raw_python_files(
Engine archive generation	backend/app/raw_python.py:1362 def engine_python_files(
Engine module pruning	backend/app/raw_python.py:1248 def _engine_module_files(
Direct support dispatch	backend/app/raw_python_support/core.py:266 async def execute(
Direct application lifecycle	backend/app/raw_python_support/main.py:24 async def run_application(
Regression suite: raw_python	backend/tests/test_raw_python.py:1 (module)
Regression suite: engine_export	backend/tests/test_engine_export.py:1 (module)

Safe change procedure

Changing runtime.py alone is insufficient when direct archives use raw_python_support. Update capability roots and generated import closure without adding unrelated helpers to every project. Validate a minimal one-activity archive, a mixed project, nested groups and a clean-environment execution. Keep vendor files and secrets out of generic archives.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Portable transformation tests and durable mediation

Responsibility

Mapper assets and message-state activities have the same implementation in Studio and both Python formats.

```
TransformationTools -> mapper suite/template/schema-impact APIs -> transformation.py
General activity -> message_state.operate -> SQLite transaction
raw_python._transformation_tools -> run_tests.py / transform_jsonl.py
```

Source locations

Responsibility	Source location / search anchor
Asset editing and test reports	frontend/src/TransformationTools.tsx:12 export default function TransformationTools
Regression suite execution	backend/app/transformation.py:69 def run_cases(
Bounded streaming batches	backend/app/transformation.py:127 def execute_batches(
Durable claim/replay/aggregation operations	backend/app/message_state.py:39 def operate(
Cooperative cancellation	backend/app/mapper.py:44 async def execute_async(
Portable test and JSONL tooling	backend/app/raw_python.py:1320 def _transformation_tools(
End-to-end capability regressions	backend/tests/test_transformation_capabilities.py:19 class TransformationCapabilityTests

Safe change procedure

Preserve fixture expectations independently of observed output. Keep checksums and template versions consistent. Test cancellation, iterator laziness, fencing tokens, restart recovery, duplicate aggregation deliveries and timeout flushing. Changes must execute in both archive formats.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Logs, faults and observable execution data

Developer guide / Features

Responsibility

Log activity output, activity-policy payload logs and persistent project log files are separate responsibilities.

```
execute_with_policy / Log activity -> runtime.log -> project_logging
API runtime-state / logs -> Studio panels
```

Source locations

Responsibility	Source location / search anchor
Runtime event emission	backend/app/runtime.py:622 def log(self,
Log activity body	backend/app/runtime.py:876 if activity.type == 'log':
Fault envelope	backend/app/runtime.py:618 def fault_payload(
Persistent log files/rotation	backend/app/project_logging.py:1 (module)
Logs API	backend/app/main.py:654 def project_logs(
Regression suite: project_logging	backend/tests/test_project_logging.py:1 (module)
Regression suite: debug_input_capture	backend/tests/test_debug_input_capture.py:1 (module)

Safe change procedure

Keep secrets and large payloads out of logs by default. Preserve timestamps, activity identity, error codes, environment path selection and rolling-file behavior. Test write permissions and failure logging as well as successful execution.

Add tests at the affected boundary, not only the UI. Regenerate this source guide after edits: anchors are checked during docs:build and participating source changes invalidate docs:check.

Every built-in function: implementation index

Where to edit

Every catalog function is listed below. Most built-ins share apply_function rather than having one Python function per name. Search the quoted function name or its aliases inside apply_function; changing the catalog does not add execution support. Runtime expression parsing and custom functions also pass through evaluate_function_expression.

Responsibility	Source location / search anchor
Implementation/aliases	backend/app/mapper.py:304 def apply_function(
Expression call parsing	backend/app/runtime.py:1639 def evaluate_function_expression(
Signatures/templates	frontend/src/mapper-functions.ts:1 (module)

Catalog index

Function / signature	Frontend definition / backend entry
concat concat(value, ...values)	frontend/src/mapper-functions.ts:13 backend/app/mapper.py :: apply_function Search literal/alias: concat
trim trim(value)	frontend/src/mapper-functions.ts:14 backend/app/mapper.py :: apply_function Search literal/alias: trim
normalizeSpace normalizeSpace(value)	frontend/src/mapper-functions.ts:15 backend/app/mapper.py :: apply_function Search literal/alias: normalizeSpace
upperCase upperCase(value)	frontend/src/mapper-functions.ts:16 backend/app/mapper.py :: apply_function Search literal/alias: upperCase
lowerCase lowerCase(value)	frontend/src/mapper-functions.ts:17 backend/app/mapper.py :: apply_function Search literal/alias: lowerCase
capitalize capitalize(value)	frontend/src/mapper-functions.ts:18 backend/app/mapper.py :: apply_function Search literal/alias: capitalize
stringLength stringLength(value)	frontend/src/mapper-functions.ts:19 backend/app/mapper.py :: apply_function Search literal/alias: stringLength
substring substring(value, start, length?)	frontend/src/mapper-functions.ts:20 backend/app/mapper.py :: apply_function Search literal/alias: substring
substringBefore substringBefore(value, token)	frontend/src/mapper-functions.ts:21 backend/app/mapper.py :: apply_function Search literal/alias: substringBefore
substringAfter substringAfter(value, token)	frontend/src/mapper-functions.ts:22 backend/app/mapper.py :: apply_function Search literal/alias: substringAfter

Function / signature	Frontend definition / backend entry
replace replace(value, pattern, replacement)	frontend/src/mapper-functions.ts:23 backend/app/mapper.py :: apply_function Search literal/alias: replace
matches matches(value, pattern)	frontend/src/mapper-functions.ts:24 backend/app/mapper.py :: apply_function Search literal/alias: matches
startsWith startsWith(value, prefix)	frontend/src/mapper-functions.ts:25 backend/app/mapper.py :: apply_function Search literal/alias: startsWith
endsWith endsWith(value, suffix)	frontend/src/mapper-functions.ts:26 backend/app/mapper.py :: apply_function Search literal/alias: endsWith
contains contains(value, token)	frontend/src/mapper-functions.ts:27 backend/app/mapper.py :: apply_function Search literal/alias: contains
compare compare(value, other)	frontend/src/mapper-functions.ts:28 backend/app/mapper.py :: apply_function Search literal/alias: compare
translate translate(value, from, to)	frontend/src/mapper-functions.ts:29 backend/app/mapper.py :: apply_function Search literal/alias: translate
padLeft padLeft(value, width, fill?)	frontend/src/mapper-functions.ts:30 backend/app/mapper.py :: apply_function Search literal/alias: padLeft
padRight padRight(value, width, fill?)	frontend/src/mapper-functions.ts:31 backend/app/mapper.py :: apply_function Search literal/alias: padRight
tokenize tokenize(value, pattern)	frontend/src/mapper-functions.ts:32 backend/app/mapper.py :: apply_function Search literal/alias: tokenize
currentDate currentDate()	frontend/src/mapper-functions.ts:34 backend/app/mapper.py :: apply_function Search literal/alias: currentDate
currentTime currentTime()	frontend/src/mapper-functions.ts:35 backend/app/mapper.py :: apply_function Search literal/alias: currentTime
currentDateTime currentDateTime()	frontend/src/mapper-functions.ts:36 backend/app/mapper.py :: apply_function Search literal/alias: currentDateTime
parseDate parseDate(value)	frontend/src/mapper-functions.ts:37 backend/app/mapper.py :: apply_function Search literal/alias: parseDate
formatDate formatDate(value, pattern)	frontend/src/mapper-functions.ts:38 backend/app/mapper.py :: apply_function Search literal/alias: formatDate
formatDateTime formatDateTime(value, pattern)	frontend/src/mapper-functions.ts:39 backend/app/mapper.py :: apply_function Search literal/alias: formatDateTime

Function / signature	Frontend definition / backend entry
addDays addDays(value, days)	frontend/src/mapper-functions.ts:40 backend/app/mapper.py :: apply_function Search literal/alias: addDays
addMonths addMonths(value, months)	frontend/src/mapper-functions.ts:41 backend/app/mapper.py :: apply_function Search literal/alias: addMonths
addHours addHours(value, hours)	frontend/src/mapper-functions.ts:42 backend/app/mapper.py :: apply_function Search literal/alias: addHours
addMinutes addMinutes(value, minutes)	frontend/src/mapper-functions.ts:43 backend/app/mapper.py :: apply_function Search literal/alias: addMinutes
addSeconds addSeconds(value, seconds)	frontend/src/mapper-functions.ts:44 backend/app/mapper.py :: apply_function Search literal/alias: addSeconds
dateDifference dateDifference(from, to)	frontend/src/mapper-functions.ts:45 backend/app/mapper.py :: apply_function Search literal/alias: dateDifference
year year(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: year
month month(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: month
day day(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: day
hour hour(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: hour
minute minute(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: minute
second second(value)	frontend/src/mapper-functions.ts:46 backend/app/mapper.py :: apply_function Search literal/alias: second
timezoneFromDateTime timezoneFromDateTime(value)	frontend/src/mapper-functions.ts:47 backend/app/mapper.py :: apply_function Search literal/alias: timezoneFromDateTime
equal equal(value, other)	frontend/src/mapper-functions.ts:49 backend/app/mapper.py :: apply_function Search literal/alias: equal
notEqual notEqual(value, other)	frontend/src/mapper-functions.ts:50 backend/app/mapper.py :: apply_function Search literal/alias: notEqual
greaterThan greaterThan(value, other)	frontend/src/mapper-functions.ts:51 backend/app/mapper.py :: apply_function Search literal/alias: greaterThan

Function / signature	Frontend definition / backend entry
lessThan lessThan(value, other)	frontend/src/mapper-functions.ts:52 backend/app/mapper.py :: apply_function Search literal/alias: lessThan
greaterOrEqual greaterOrEqual(value, other)	frontend/src/mapper-functions.ts:53 backend/app/mapper.py :: apply_function Search literal/alias: greaterOrEqual
lessOrEqual lessOrEqual(value, other)	frontend/src/mapper-functions.ts:54 backend/app/mapper.py :: apply_function Search literal/alias: lessOrEqual
add add(value, other)	frontend/src/mapper-functions.ts:55 backend/app/mapper.py :: apply_function Search literal/alias: add
subtract subtract(value, other)	frontend/src/mapper-functions.ts:56 backend/app/mapper.py :: apply_function Search literal/alias: subtract
multiply multiply(value, other)	frontend/src/mapper-functions.ts:57 backend/app/mapper.py :: apply_function Search literal/alias: multiply
divide divide(value, other)	frontend/src/mapper-functions.ts:58 backend/app/mapper.py :: apply_function Search literal/alias: divide
mod mod(value, other)	frontend/src/mapper-functions.ts:59 backend/app/mapper.py :: apply_function Search literal/alias: mod
number number(value)	frontend/src/mapper-functions.ts:61 backend/app/mapper.py :: apply_function Search literal/alias: number
integer integer(value)	frontend/src/mapper-functions.ts:62 backend/app/mapper.py :: apply_function Search literal/alias: integer
round round(value, precision?)	frontend/src/mapper-functions.ts:63 backend/app/mapper.py :: apply_function Search literal/alias: round
roundHalfToEven roundHalfToEven(value, precision?)	frontend/src/mapper-functions.ts:64 backend/app/mapper.py :: apply_function Search literal/alias: roundHalfToEven
floor floor(value)	frontend/src/mapper-functions.ts:65 backend/app/mapper.py :: apply_function Search literal/alias: floor
ceiling ceiling(value)	frontend/src/mapper-functions.ts:66 backend/app/mapper.py :: apply_function Search literal/alias: ceiling
abs abs(value)	frontend/src/mapper-functions.ts:67 backend/app/mapper.py :: apply_function Search literal/alias: abs
sqrt sqrt(value)	frontend/src/mapper-functions.ts:68 backend/app/mapper.py :: apply_function Search literal/alias: sqrt

Function / signature	Frontend definition / backend entry
power power(value, exponent)	frontend/src/mapper-functions.ts:69 backend/app/mapper.py :: apply_function Search literal/alias: power
modulo modulo(value, divisor)	frontend/src/mapper-functions.ts:70 backend/app/mapper.py :: apply_function Search literal/alias: modulo
clamp clamp(value, minimum, maximum)	frontend/src/mapper-functions.ts:71 backend/app/mapper.py :: apply_function Search literal/alias: clamp
min min(values)	frontend/src/mapper-functions.ts:72 backend/app/mapper.py :: apply_function Search literal/alias: min
max max(values)	frontend/src/mapper-functions.ts:72 backend/app/mapper.py :: apply_function Search literal/alias: max
sum sum(values)	frontend/src/mapper-functions.ts:72 backend/app/mapper.py :: apply_function Search literal/alias: sum
average average(values)	frontend/src/mapper-functions.ts:72 backend/app/mapper.py :: apply_function Search literal/alias: average
count count(values)	frontend/src/mapper-functions.ts:72 backend/app/mapper.py :: apply_function Search literal/alias: count
lookup lookup(value, table, default?)	frontend/src/mapper-functions.ts:74 backend/app/mapper.py :: apply_function Search literal/alias: lookup
lookupTable lookupTable(value, name, default?)	frontend/src/mapper-functions.ts:75 backend/app/mapper.py :: apply_function Search literal/alias: lookupTable
enrich enrich(record, fields)	frontend/src/mapper-functions.ts:76 backend/app/mapper.py :: apply_function Search literal/alias: enrich
chunks chunks(records, size)	frontend/src/mapper-functions.ts:77 backend/app/mapper.py :: apply_function Search literal/alias: chunks
joinBy joinBy(left, right, leftKey, rightKey)	frontend/src/mapper-functions.ts:78 backend/app/mapper.py :: apply_function Search literal/alias: joinBy
group group(values, path?)	frontend/src/mapper-functions.ts:79 backend/app/mapper.py :: apply_function Search literal/alias: group
top top(values, count?)	frontend/src/mapper-functions.ts:80 backend/app/mapper.py :: apply_function Search literal/alias: top
distinctValues distinctValues(values)	frontend/src/mapper-functions.ts:81 backend/app/mapper.py :: apply_function Search literal/alias: distinctValues

Function / signature	Frontend definition / backend entry
sort sort(values, descending?)	frontend/src/mapper-functions.ts:82 backend/app/mapper.py :: apply_function Search literal/alias: sort
reverse reverse(values)	frontend/src/mapper-functions.ts:83 backend/app/mapper.py :: apply_function Search literal/alias: reverse
first first(values)	frontend/src/mapper-functions.ts:84 backend/app/mapper.py :: apply_function Search literal/alias: first
last last(values)	frontend/src/mapper-functions.ts:85 backend/app/mapper.py :: apply_function Search literal/alias: last
indexOf indexOf(values, item)	frontend/src/mapper-functions.ts:86 backend/app/mapper.py :: apply_function Search literal/alias: indexOf
subsequence subsequence(values, start, end?)	frontend/src/mapper-functions.ts:87 backend/app/mapper.py :: apply_function Search literal/alias: subsequence
flatten flatten(values)	frontend/src/mapper-functions.ts:88 backend/app/mapper.py :: apply_function Search literal/alias: flatten
join join(values, separator)	frontend/src/mapper-functions.ts:89 backend/app/mapper.py :: apply_function Search literal/alias: join
split split(value, pattern)	frontend/src/mapper-functions.ts:90 backend/app/mapper.py :: apply_function Search literal/alias: split
deepEqual deepEqual(value, other)	frontend/src/mapper-functions.ts:91 backend/app/mapper.py :: apply_function Search literal/alias: deepEqual
string string(value)	frontend/src/mapper-functions.ts:93 backend/app/mapper.py :: apply_function Search literal/alias: string
boolean boolean(value)	frontend/src/mapper-functions.ts:94 backend/app/mapper.py :: apply_function Search literal/alias: boolean
jsonParse jsonParse(value)	frontend/src/mapper-functions.ts:95 backend/app/mapper.py :: apply_function Search literal/alias: jsonParse
jsonRender jsonRender(value)	frontend/src/mapper-functions.ts:96 backend/app/mapper.py :: apply_function Search literal/alias: jsonRender
render-xml render-xml(value, rootName?, pretty?)	frontend/src/mapper-functions.ts:97 backend/app/mapper.py :: apply_function Search literal/alias: render-xml
coalesce coalesce(value, ...fallbacks)	frontend/src/mapper-functions.ts:99 backend/app/mapper.py :: apply_function Search literal/alias: coalesce

Function / signature	Frontend definition / backend entry
default default(value, fallback)	frontend/src/mapper-functions.ts:100 backend/app/mapper.py :: apply_function Search literal/alias: default
exists exists(value)	frontend/src/mapper-functions.ts:101 backend/app/mapper.py :: apply_function Search literal/alias: exists
empty empty(value)	frontend/src/mapper-functions.ts:102 backend/app/mapper.py :: apply_function Search literal/alias: empty
not not(value)	frontend/src/mapper-functions.ts:103 backend/app/mapper.py :: apply_function Search literal/alias: not
ifThenElse ifThenElse(condition, trueValue, falseValue)	frontend/src/mapper-functions.ts:104 backend/app/mapper.py :: apply_function Search literal/alias: ifThenElse
uuid uuid()	frontend/src/mapper-functions.ts:105 backend/app/mapper.py :: apply_function Search literal/alias: uuid
base64Encode base64Encode(value)	frontend/src/mapper-functions.ts:107 backend/app/mapper.py :: apply_function Search literal/alias: base64Encode
base64Decode base64Decode(value)	frontend/src/mapper-functions.ts:108 backend/app/mapper.py :: apply_function Search literal/alias: base64Decode
hexEncode hexEncode(value)	frontend/src/mapper-functions.ts:109 backend/app/mapper.py :: apply_function Search literal/alias: hexEncode
hexDecode hexDecode(value)	frontend/src/mapper-functions.ts:110 backend/app/mapper.py :: apply_function Search literal/alias: hexDecode
urlEncode urlEncode(value)	frontend/src/mapper-functions.ts:111 backend/app/mapper.py :: apply_function Search literal/alias: urlEncode
urlDecode urlDecode(value)	frontend/src/mapper-functions.ts:112 backend/app/mapper.py :: apply_function Search literal/alias: urlDecode
hash hash(value, algorithm?)	frontend/src/mapper-functions.ts:113 backend/app/mapper.py :: apply_function Search literal/alias: hash